Home Assignment

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1) Definitions:

verification: process of evaluating work-products to check whether they meet specified requirements. Are we building the product right

validation: process of evaluating the final product to ensure it meets user needs. Are we building the right product.

Error: Human mistake made by developer.

Bug: Informal term for a defect in software.

fault: Incorrect step, process or data definition in a program.

Defect: Mismatch between expected and actual result.

Testing: process of executing software to identify defects.

Debugging: process of finding and fixing defects.

failure: when software does not perform expected function.

Severity: Impact of defect on system functionality.

Priority: urgency of fixing the defect (High, medium low).

2. Phases of STLC (Software Testing Life cycle).

1. Requirement Analysis - understand requirement and identify testable aspects.
2. Test planning - Define test strategy, scope, resources.
3. Test case Development - Create test cases and test data.
4. Test Environment setup - prepare hardware/software setup
5. Test Execution - Execute test cases and log defects
6. Test cycle closure - Evaluate test coverage, prepare reports.

Seven principles of Software Testing:

1. Testing shows presence of defects.
2. Exhaustive testing is impossible.
3. Early testing saves time and cost.
4. Defect clustering.
5. pesticide paradox.
6. Testing is context dependent.
7. Absence of errors is a fallacy.

3. a) Agile Development methodology

* Iterative and incremental model.
* continuous testing.
* Customer collaboration.
* Example : Scrum.

b) Traditional Development process.

waterfall Model

* sequential phases.
* Requirements -> Design -> Development -> Testing -> maintenance.

Spiral Model

* Risk-driven model.
* Combines iterative development with risk analysis.

4. Objectives of Software Testing

* Detect defects.
* Ensure quality.
* validate requirements.
* Improve reliability.
* Reduce maintenance cost.

Importance of Test metrics

* measure quality.
* Track defect density.

- Monitor test coverage.
- Evaluate team performance.
- Help decision-making.

5) User Stories (E-Commerce Application)

User story format:

As a [user], I want [feature] so that [benefit].

1. As a customer, I want to register an account so that I can place orders.
2. As a customer, I want to search product so that I can find items easily.
3. As a customer, I want to add items to cart so that I can purchase later.
4. As a customer, I want secure payment options so that my transaction is safe.
5. As a admin, I want to manage product inventory so that stock is updated.